



FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

INTEGRATIVE PROGRAMMING AND TECHNOLOGIES III

TEST 1 Memo

DATE: 20 MARCH 2024

TIME: 1 Hour

EXAMINER: BOAMAH-ABU, CHARLES (PhD) **MARKS:** 40

NO. OF PAGES: 5 (Including cover page)

INSTRUCTIONS/REQUIREMENTS

1. This examination is closed book
2. Answer all questions
3. Where stated "briefly", please give a short answer
4. Ensure your handwriting is readable

Question 1 7 marks

In each of the following, indicate whether the statement is **true** or **false**. Write only **T** (for True) and **F** (for False).

1. Both **abstract classes** and **interfaces** can contain constant values.
2. Integrative programming allows us to integrate **only** components of the same platform.
3. Integrative programming **requires** that some of the components to be integrated exist.
4. An interface is a special type of abstract class.
5. It is easier to maintain classes which are independent but work together.
6. Always design classes that strongly depend on each other.
7. Code reuse increases the total cost of software development.

Answer: T, F, T, T, T, F, F

Question 2 8 marks

Briefly answer the following:

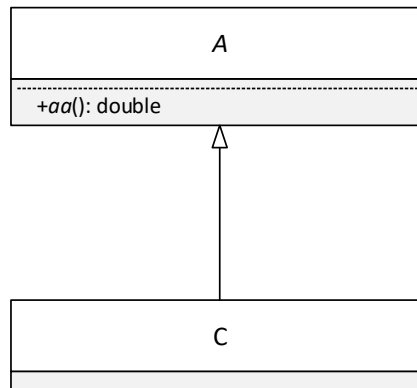
1. Give one reason why Integrative programming is important. (2)
2. Name the two main types of relationships in object oriented programming. (2)
3. What is the main difference between an abstract class and interface? (2)
4. Give one benefit of abstract classes. (2)

Answer: any correct answer

1. Inheritance✓ and composition✓
2. Abstract class can contain method implementation; interface cannot. ✓✓
3. Integrative programming allows integration of different organizational resources. ✓✓
4. Abstract classes support code reuse ✓✓

Question 3 14 marks

3.1 The diagram below shows class **A** contains an abstract method and class **C** inherits **A**.



a. What is wrong with the code segments below. (2)

```
public class A {
    ...
    public abstract double aa();
    ...
}
```

b. What is wrong with the code segments below. (3)

```
public class C{
    ...
    double aa() {
    ...
    }
}
```

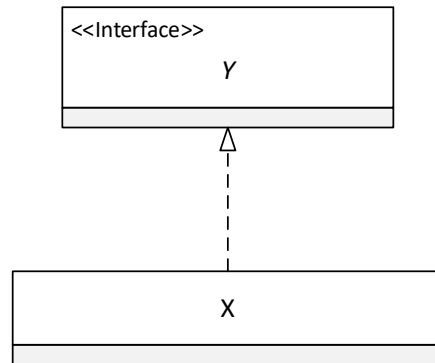
Answer:

a. Incorrect header: **A** must be declared as abstract. ✓✓

b. i. **B** extends ✓

ii. Methods in an interface are, by default, public, so the access level is reduced. ✓✓

- 3.2. **X** is a concrete class and **Y** is an interface. In each of the following, state whether the instantiation is valid. Explain your answer. (6)



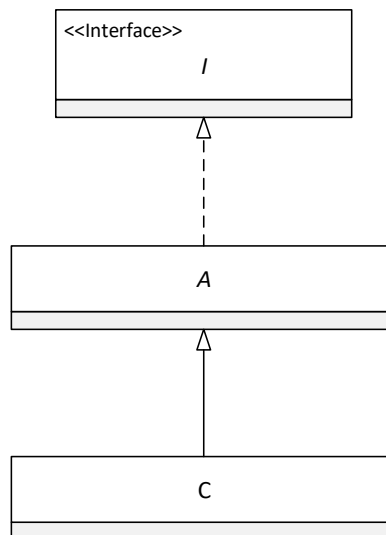
- a. Y y1=new Y();
- b. Y y2=new X();
- c. X x1=new X();

Answer:

- a. Invalid: Y is an interface, cannot create an instance ✓✓
- b. Valid: a variable of an interface can hold instances of a class that implements it ✓✓
- c. Valid: can create an instance of a concrete class ✓✓

- 3.3. The diagram below shows an abstract class **A** implements the interface **I** and the class **C** inherits **A**.

- a. Write the class structure for A. (3)
- b. Write the class structure for C. (2)

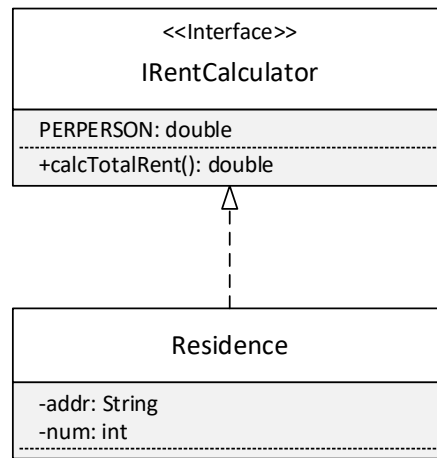


Answer:

- a. public abstract ✓class A implements I ✓✓
- b. public class C extends A ✓✓

Question 4 10 marks

Ngcwele Student Residences provides shared accommodation to student. The diagram below shows the relationship between the class **Residence** and the interface **IRentCalculator**.



Residence contains **addr** (the address of a residence) and **num** (number of students in the residence). **IRentCalculator** contains **PERPERSON** a constant (R2100.00, the rent paid by each student in the residence). The method **calcTotalRent()** returns the total rent paid by all students in a residence.

4.1 Write the code for **IRentCalculator**. (5)

4.2 Write the code for **calcTotalRent()** in **Residence**. (4)

Answer:

4.1

```
public interface RentCalculator ✓{
    //double PERPERSON=2100; also correct
    public static final double PERPERSON=2100; ✓✓
    public abstract double calcTotalRental(); ✓✓
}
```

4.2

```
public double calcTotalRental() ✓✓{
    return PERPERSON * num; ✓✓
}
```